

Rafay Ahmad Raza

rafayahmadraza@gmail.com

| <https://github.com/RafayAhmadRaza>

| <https://www.linkedin.com/in/rafay-ahmad-raza/>

| +923301587838

Education

National Skills University

B.S. Computer Science

Faiz Ahmed Faiz Road, Sector H-8/1, 44310

Islamabad

18/09/2022 - 20/08/2026

- GPA: 3.48/4.0
- Relevant Coursework: Software Engineering, Operating Systems, Analysis of Algorithms, Artificial Intelligence, Machine Learning, Data structures and Algorithms.

Experience

Pakistan Telecommunication Corporation

Intern - Software Development

P3MR+G9M, Constitution Ave, F-5/1 F-5,

Islamabad, 44000, Pakistan

19/06/2025 - 18/08/2025

- Architected a full-stack web application by integrating a custom **TensorFlow** model with a **Python** backend and **Bootstrap** frontend.
- Developed and utilized **REST APIs** to facilitate seamless data communication between the web interface and machine learning models.
- Engineered image capture capabilities to allow real-time facial analysis through **AI-driven** processing of user uploads.
- Applied clean code principles, ensuring **modular design** and industry-standard naming conventions for enhanced scalability.

Projects

Original Classifier 151 Pokemon Image

Developer

24/11/2024 - 06/01/2025

- Built an image classification model to identify the original 151 Pokémon using **TensorFlow** and **Python**.
- Preprocessed and analyzed dataset using **NumPy**, **Pandas**, and **Matplotlib**; applied normalization techniques for model training.
- Developed and trained a custom **Neural Network** for image recognition tasks.
- Created a **Streamlit web interface** for interactive testing and predictions.

Glasses Recommender System

Developer

24/07/2025 - 28/07/2025

- Built a face shape recognition system using **TensorFlow** and **Python** to recommend suitable eyewear styles.
- Preprocessed and analyzed datasets using **NumPy**, **Pandas**, and **Matplotlib**; utilized **MediaPipe** for facial landmark extraction and applied normalization techniques for model training.
- Developed a web interface using **Bootstrap**, **HTML**, **CSS**, and **JavaScript** with a **FastAPI** and **uvicorn** backend.
- Secured the **Gemini API** key via environment variables and integrated it to generate personalized eyewear recommendations based on predicted face shape.

Sign Buddy (FYP - Expected Completion: June 2026)

Developer

- Architecting a cross-platform mobile application for real-time sign language translation utilizing the **Flutter** framework.
- Executed a literature review of existing sign language recognition solutions to optimize model architecture.
- Engineering both finger-spell and word-level sign language recognition models via **TensorFlow**.
- Preprocessed and analyzed datasets using **NumPy, Pandas, Matplotlib, and Seaborn**; applied normalization techniques for model training.
- Integrated **Firestore Authentication** to facilitate secure user account management and data security.
- Authored comprehensive technical documentation detailing system architecture and modular application design.

Certifications & Assessments

- **National Skills Competency Skills (NSCT)** - Achieved 89.7th Percentile Nationwide.
- **Coursera:** Coursera Specialization in Machine Learning (2025),
 - Supervised Machine Learning: Regression and Classification,
 - Advanced Learning Algorithms,
 - Unsupervised Learning, Recommenders, Reinforcement Learning'Coursera Certificate in Object Oriented Programming in Java (2023)
Coursera Certificate in Programming for Everybody (Getting Started with Python) (2025)

Technical Skills

- **Programming Languages:** Python, Dart
- **Web Technologies:** HTML, CSS, JavaScript (*Basic*)
- **Languages:** Urdu (Native), English (Fluent), German (A1)
- **Systems:** Linux, Windows.
- **Tools & Technologies:** TensorFlow, MediaPipe, Firebase, FastAPI, Streamlit, Gemini API,Flutter